

CS257 Linear and Convex Optimization

Homework 3

Due: October 12, 2020

September 29, 2020

1. Prove that $\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_m \in \mathbb{R}^n$ are affinely independent if and only if

$$\sum_{i=0}^m c_i \mathbf{x}_i = \mathbf{0} \text{ and } \sum_{i=0}^m c_i = 0 \implies c_i = 0 \text{ for } i = 0, 1, \dots, m. \quad (\star)$$

Proof. “ $\Rightarrow$ ”. Assume $\mathbf{x}_0, \dots, \mathbf{x}_m$ are affinely independent. We need to show $(\star)$. Suppose $\sum_{i=0}^m c_i \mathbf{x}_i = \mathbf{0}$ and $\sum_{i=0}^m c_i = 0$. Note

$$c_0 = -\sum_{i=1}^m c_i.$$

So

$$\mathbf{0} = \sum_{i=0}^m c_i \mathbf{x}_i = \left(-\sum_{i=1}^m c_i \right) \mathbf{x}_0 + \sum_{i=1}^m c_i \mathbf{x}_i = \sum_{i=1}^m c_i (\mathbf{x}_i - \mathbf{x}_0)$$

Since $\mathbf{x}_0, \dots, \mathbf{x}_m$ are affinely independent, $\mathbf{x}_1 - \mathbf{x}_0, \dots, \mathbf{x}_m - \mathbf{x}_0$ are linearly independent, and hence $c_i = 0$ for $i = 1, \dots, m$. Then $c_0 = -\sum_{i=1}^m c_i = 0$. So $(\star)$ holds.

“ $\Leftarrow$ ”. Assume $(\star)$ is true. We need to show $\mathbf{x}_1 - \mathbf{x}_0, \dots, \mathbf{x}_m - \mathbf{x}_0$ are linearly independent. Suppose

$$\sum_{i=1}^m c_i (\mathbf{x}_i - \mathbf{x}_0) = \mathbf{0}.$$

Let $c_0 = -\sum_{i=1}^m c_i$. Then $\sum_{i=0}^m c_i = 0$, and

$$\mathbf{0} = \sum_{i=1}^m c_i (\mathbf{x}_i - \mathbf{x}_0) = \sum_{i=1}^m c_i \mathbf{x}_i - \sum_{i=1}^m c_i \mathbf{x}_0 = \sum_{i=0}^m c_i \mathbf{x}_i$$

By $(\star)$, $c_i = 0$ for $i = 0, 1, \dots, m$, so $\mathbf{x}_1 - \mathbf{x}_0, \dots, \mathbf{x}_m - \mathbf{x}_0$ are linearly independent. $\square$

2. Given two probability distributions $\mathbf{x}, \mathbf{y} \in \Delta_{n-1}$, where Δ_{n-1} is the probability simplex, the **Kullback-Leibler (KL) divergence** between them is defined by

$$KL(\mathbf{x} \parallel \mathbf{y}) = \sum_{i=1}^n x_i \log \frac{x_i}{y_i}$$

Use the concavity of log to show that $KL(\mathbf{x} \parallel \mathbf{y}) \geq 0$. You can assume $\mathbf{x} > \mathbf{0}, \mathbf{y} > \mathbf{0}$.

Proof. Note that $-\log x$ is strictly convex.

$$KL(\mathbf{x} \parallel \mathbf{y}) = -\sum_{i=1}^n x_i \log \frac{y_i}{x_i} \geq -\log \left(\sum_{i=1}^n x_i \frac{y_i}{x_i} \right) = -\log \left(\sum_{i=1}^n y_i \right) = -\log 1 = 0$$

□

3. Prove that the set of global minima of a convex function over a convex set is convex, i.e. if f is convex and $S \subset \text{dom } f$ is convex, then the set M defined below is also convex

$$M = \{\mathbf{x}^* \in S : f(\mathbf{x}^*) \leq f(\mathbf{x}), \forall \mathbf{x} \in S\}$$

Proof. Let $\mathbf{x}^*, \mathbf{y}^* \in M$ and $\theta \in [0, 1]$. Since S is convex, $\theta \mathbf{x}^* + \bar{\theta} \mathbf{y}^* \in S$. Since f is convex, for any $\mathbf{x} \in S$,

$$f(\theta \mathbf{x}^* + \bar{\theta} \mathbf{y}^*) \leq \theta f(\mathbf{x}^*) + \bar{\theta} f(\mathbf{y}^*) \leq \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{x}) = f(\mathbf{x})$$

Thus $\theta \mathbf{x}^* + \bar{\theta} \mathbf{y}^* \in M$ and M is convex.

□

4. Let f be convex. If $f(\theta \mathbf{x} + \bar{\theta} \mathbf{y}) = \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{y})$ for some $\mathbf{x}, \mathbf{y}$ and $\theta = \theta_0 \in (0, 1)$, then it holds for the same $\mathbf{x}, \mathbf{y}$ and any $\theta \in [0, 1]$.

Hint: Assume $f(\theta_1 \mathbf{x} + \bar{\theta}_1 \mathbf{y}) < \theta_1 f(\mathbf{x}) + \bar{\theta}_1 f(\mathbf{y})$ for some θ_1 . Without loss of generality, you may assume $\theta_1 \in (0, \theta_0)$; the case $\theta_1 \in (\theta_0, 1)$ is similar. Express $\theta_0 \mathbf{x} + \bar{\theta}_0 \mathbf{y}$ as a convex combination of $\theta_1 \mathbf{x} + \bar{\theta}_1 \mathbf{y}$ and $\mathbf{x}$. Then deduce a contradiction.

Proof. Suppose $f(\theta_1 \mathbf{x} + \bar{\theta}_1 \mathbf{y}) < \theta_1 f(\mathbf{x}) + \bar{\theta}_1 f(\mathbf{y})$ for some $\theta_1 \in (0, \theta_0)$. Let $\mathbf{z}_i = \theta_i \mathbf{x} + \bar{\theta}_i \mathbf{y}$ for $i = 0, 1$. We want to express $\mathbf{z}_0$ as a convex combination of $\mathbf{x}$ and $\mathbf{z}_1$, i.e. we want to find α s.t.

$$\alpha \mathbf{x} + \bar{\alpha} \mathbf{z}_1 = \alpha \mathbf{x} + \bar{\alpha}(\theta_1 \mathbf{x} + \bar{\theta}_1 \mathbf{y}) = \mathbf{z}_0 = \theta_0 \mathbf{x} + \bar{\theta}_0 \mathbf{y}$$

Comparing the coefficients of $\mathbf{x}$ and $\mathbf{y}$, we see it suffices to set $\bar{\alpha} = \bar{\theta}_0 / \bar{\theta}_1 \in (0, 1)$. Thus

$$\begin{aligned} f(\mathbf{z}_0) &= f(\alpha \mathbf{x} + \bar{\alpha} \mathbf{z}_1) \\ &\leq \alpha f(\mathbf{x}) + \bar{\alpha} f(\mathbf{z}_1) \quad \text{by convexity of } f \\ &< \alpha f(\mathbf{x}) + \bar{\alpha}[\theta_1 f(\mathbf{x}) + \bar{\theta}_1 f(\mathbf{y})] \quad \text{by assumption about } \mathbf{z}_1 \\ &= \theta_0 f(\mathbf{x}) + \bar{\theta}_0 f(\mathbf{y}) \quad \text{by definition of } \alpha \\ &= f(\mathbf{z}_0) \quad \text{by assumption about } \mathbf{z}_0 \end{aligned}$$

which is a contradiction, so $f(\theta \mathbf{x} + \bar{\theta} \mathbf{y}) = \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{y})$ for all $\theta \in (0, \theta_0)$. A similar proof shows $f(\theta \mathbf{x} + \bar{\theta} \mathbf{y}) = \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{y})$ for $\theta \in (\theta_0, 1)$. □

5. Suppose $f : S \subset \mathbb{R} \rightarrow \mathbb{R}$ is convex. Let $a, b \in S$ and $a < b$.

(a). Show

$$f(x) \leq \frac{b-x}{b-a} f(a) + \frac{x-a}{b-a} f(b), \quad \forall x \in [a, b]$$

(b). Show

$$\frac{f(x) - f(a)}{x - a} \leq \frac{f(b) - f(a)}{b - a} \leq \frac{f(b) - f(x)}{b - x}, \quad \forall x \in (a, b)$$

What's the geometric interpretation of these inequalities? Draw a sketch to illustrate them.

(c). Suppose f is differentiable. Use (b) to show

$$f'(a) \leq \frac{f(b) - f(a)}{b - a} \leq f'(b)$$

What can you say about the monotonicity of the derivative f' ?

(d). Suppose f is twice differentiable. Note $a < b$ are arbitrary points in S . Use (c) to show that $f''(x) \geq 0$.

Proof. (a). Note

$$x = \frac{b-x}{b-a}a + \frac{x-a}{b-a}b$$

Since f is convex, Jensen's inequality implies

$$f(x) \leq \frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b)$$

(b). By (a)

$$f(x) - f(a) \leq \frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b) - f(a) = \frac{x-a}{b-a}[f(b) - f(a)]$$

so

$$\frac{f(x) - f(a)}{x - a} \leq \frac{f(b) - f(a)}{b - a}$$

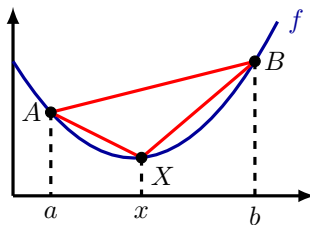
Similarly,

$$f(b) - f(x) \geq f(b) - \frac{b-x}{b-a}f(a) - \frac{x-a}{b-a}f(b) - f(a) = \frac{b-x}{b-a}[f(b) - f(a)]$$

so

$$\frac{f(b) - f(a)}{b - a} \leq \frac{f(b) - f(x)}{b - x}$$

Let $A = (a, f(a))$, $B = (b, f(b))$, $X = (x, f(x))$. The inequalities assert that the slope of the line AX is no larger than that of the line AB , which in turn is no larger than that of XB .



(c). Letting $x \rightarrow a$ in the first inequality of (b),

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \leq \frac{f(b) - f(a)}{b - a}$$

Letting $x \rightarrow b$ in the second inequality of (b),

$$\frac{f(b) - f(a)}{b - a} \leq \lim_{x \rightarrow b} \frac{f(b) - f(x)}{b - x} = f'(b)$$

f' is monotonically increasing.

(d). By (c), $f'(x) \leq f'(y)$ for any $y > x$, so

$$\frac{f'(y) - f'(x)}{y - x} \geq 0$$

Letting $y \rightarrow x$,

$$f''(x) = \lim_{y \rightarrow x} \frac{f'(y) - f'(x)}{y - x} \geq 0$$

□

6. Let $f : (a, b) \rightarrow \mathbb{R}$ be convex, where $-\infty \leq a < b \leq +\infty$. Let X be random variable taking values in (a, b) . Suppose the expectations $\mathbb{E}X$ and $\mathbb{E}f(X)$ exist. Prove Jensen's inequality $f(\mathbb{E}X) \leq \mathbb{E}f(X)$ by completing the following steps.

(a). Let $\mu = \mathbb{E}X$. Show that there exists $\beta \in \mathbb{R}$ such that

$$f(x) \geq f(\mu) + \beta(x - \mu), \quad \forall x \in (a, b) \quad (\star)$$

Hint: You can take

$$\beta = \sup_{a < s < \mu} \frac{f(\mu) - f(s)}{\mu - s}.$$

Obviously $\beta > -\infty$. Use part (b) of Problem 5 to show that $\beta < +\infty$ and satisfies $(\star)$ (consider $a < x < \mu$ and $\mu < x < b$ separately).

(b). Show that

$$f(X) \geq f(\mu) + \beta(X - \mu).$$

and conclude $\mathbb{E}f(X) \geq f(\mathbb{E}X)$ by taking expectation.

Remark. If f is differentiable, we can take $\beta = f'(\mu)$ by the first-order condition. Part (a) shows that $(\star)$ holds without assuming differentiability. The number β used in the proof generalizes the concept of gradient (derivative) $f'(\mu)$. Any β satisfying $(\star)$ is called a **subgradient** of f at μ . For example, any $\beta \in [-1, 1]$ is a subgradient of $f(x) = |x|$ at 0.

Proof. (a). By part (b) of Problem 5, for $a < s < \mu < u < b$,

$$\frac{f(\mu) - f(s)}{\mu - s} \leq \frac{f(u) - f(\mu)}{u - \mu},$$

so

$$\beta = \sup_{a < s < \mu} \frac{f(\mu) - f(s)}{\mu - s} \leq \frac{f(u) - f(\mu)}{u - \mu} < +\infty.$$

Thus

$$\frac{f(\mu) - f(s)}{\mu - s} \leq \beta \leq \frac{f(u) - f(\mu)}{u - \mu}$$

Rearranging,

$$f(s) \geq f(\mu) + \beta(s - \mu), \quad f(u) \geq f(\mu) + \beta(u - \mu)$$

Therefore,

$$f(x) \geq f(\mu) + \beta(x - \mu), \quad \forall x \in (a, b)$$

(b). Since $X \in (a, b)$, by (a)

$$f(X) \geq f(\mu) + \beta(X - \mu).$$

Taking expectation, we obtain

$$\mathbb{E}f(X) \geq \mathbb{E}[f(\mu) + \beta(X - \mu)] = f(\mu) + \beta(\mathbb{E}X - \mu) = f(\mu) = f(\mathbb{E}X)$$

since $\mu = \mathbb{E}X$.

□

7. In this problem, we show that the so-called **midpoint convexity** and continuity implies convexity. Assume $f : \mathbb{R} \rightarrow \mathbb{R}$ is **midpoint convex**, i.e.

$$f\left(\frac{x_1 + x_2}{2}\right) \leq \frac{1}{2}[f(x_1) + f(x_2)], \quad \forall x_1, x_2 \in \mathbb{R}.$$

(a). Show that for $k \in \mathbb{N}$,

$$f\left(\frac{x_1 + x_2 + \cdots + x_{2^k}}{2^k}\right) \leq \frac{1}{2^k}[f(x_1) + f(x_2) + \cdots + f(x_{2^k})], \quad \forall x_1, x_2, \dots, x_{2^k} \in \mathbb{R}.$$

Hint: Use induction on k .

(b). Show that for $n \in \mathbb{N}$,

$$f\left(\frac{x_1 + x_2 + \cdots + x_n}{n}\right) \leq \frac{1}{n}[f(x_1) + f(x_2) + \cdots + f(x_n)], \quad \forall x_1, x_2, \dots, x_n \in \mathbb{R}.$$

Hint: Assume $2^{k-1} < n \leq 2^k$. Let $x_i = \bar{x} = \frac{x_1 + \cdots + x_n}{n}$ for $i = n+1, \dots, 2^k$ in part (a).

(c). Use (b) to show that for $p, q \in \mathbb{N}$,

$$f\left(\frac{p}{p+q}x_1 + \frac{q}{p+q}x_2\right) \leq \frac{p}{p+q}f(x_1) + \frac{q}{p+q}f(x_2), \quad \forall x_1, x_2 \in \mathbb{R}.$$

Conclude that for rational $\theta \in \mathbb{Q} \cap (0, 1)$,

$$f(\theta x_1 + \bar{\theta} x_2) \leq \theta f(x_1) + \bar{\theta} f(x_2), \quad \forall x_1, x_2 \in \mathbb{R}.$$

(d). Assume f is also continuous. Show that for $\theta \in [0, 1]$,

$$f(\theta x_1 + \bar{\theta} x_2) \leq \theta f(x_1) + \bar{\theta} f(x_2), \quad \forall x_1, x_2 \in \mathbb{R}.$$

Hint: Let $\{\theta_n\}$ be a sequence of rational numbers in $\mathbb{Q} \cap (0, 1)$ such that $\theta_n \rightarrow \theta$ as $n \rightarrow \infty$.

Proof. (a). Use induction on k . The base case $k = 1$ holds by assumption. Assume the equality holds for $k - 1$.

$$\begin{aligned} f\left(\frac{x_1 + x_2 + \cdots + x_{2^k}}{2^k}\right) &= f\left(\frac{\frac{x_1+x_2}{2} + \frac{x_3+x_4}{2} + \cdots + \frac{x_{2^{k-1}}+x_{2^k}}{2}}{2^{k-1}}\right) \\ &\leq \frac{1}{2^{k-1}} \left[f\left(\frac{x_1+x_2}{2}\right) + f\left(\frac{x_3+x_4}{2}\right) + \cdots + f\left(\frac{x_{2^{k-1}}+x_{2^k}}{2}\right) \right] \\ &\leq \frac{1}{2^k} [f(x_1) + f(x_2) + \cdots + f(x_{2^k})]. \end{aligned}$$

(b). Assume $2^{k-1} < n \leq 2^k$. Setting $x_i = \bar{x} = \frac{x_1 + \cdots + x_n}{n}$ for $i = n+1, \dots, 2^k$ in part (a),

$$f(\bar{x}) = f\left(\frac{x_1 + x_2 + \cdots + x_n + (2^k - n)\bar{x}}{2^k}\right) \leq \frac{1}{2^k} [f(x_1) + f(x_2) + \cdots + f(x_n) + (2^k - n)f(\bar{x})].$$

By rearranging,

$$f(\bar{x}) = f\left(\frac{x_1 + x_2 + \cdots + x_n}{n}\right) \leq \frac{1}{n} [f(x_1) + f(x_2) + \cdots + f(x_n)].$$

(c). Let $n = p + q$. By part (b),

$$\begin{aligned} f\left(\frac{p}{p+q}x_1 + \frac{q}{p+q}x_2\right) &= f\left(\frac{\overbrace{x_1 + \cdots + x_1}^{p \text{ times}} + \overbrace{x_2 + \cdots + x_2}^{q \text{ times}}}{p+q}\right) \\ &\leq \frac{1}{p+q} [\overbrace{f(x_1) + \cdots + f(x_1)}^{p \text{ times}} + \overbrace{f(x_2) + \cdots + f(x_2)}^{q \text{ times}}] \\ &= \frac{p}{p+q} f(x_1) + \frac{q}{p+q} f(x_2). \end{aligned}$$

Since any rational $\theta \in \mathbb{Q} \cap (0, 1)$ can be represented as $p/(p+q)$ for some $p, q \in \mathbb{N}$, we have

$$f(\theta x_1 + \bar{\theta} x_2) \leq \theta f(x_1) + \bar{\theta} f(x_2), \quad \forall x_1, x_2 \in \mathbb{R}, \theta \in \mathbb{Q} \cap (0, 1)$$

(d). For $\theta \in [0, 1]$, there exists a sequence of rational numbers $\theta_n \in \mathbb{Q} \cap (0, 1)$ such that $\theta_n \rightarrow \theta$ as $n \rightarrow \infty$.

By part (c),

$$f(\theta_n x_1 + \bar{\theta}_n x_2) \leq \theta_n f(x_1) + \bar{\theta}_n f(x_2).$$

Letting $n \rightarrow \infty$ and using the continuity of f ,

$$f(\theta x_1 + \bar{\theta} x_2) = \lim_{n \rightarrow \infty} f(\theta_n x_1 + \bar{\theta}_n x_2) \leq \lim_{n \rightarrow \infty} [\theta_n f(x_1) + \bar{\theta}_n f(x_2)] = \theta f(x_1) + \bar{\theta} f(x_2).$$

□